Answers on
Booklet - 2

Taken from OCR Physics A past papers

Abbreviations, annotations and conventions used in the Mark Scheme	/ = alternative and acceptable answers for the same marking point ; = separates marking points NOT = answers which are not worthy of credit ( ) = words which are not essential to gain credit ____ = (underlining) key words which **must** be used to gain credit ecf = error carried forward

Booklet 2 Answers

1.

(a)	weight = 60 x 9.81 (allow 9.8 but not 10) = 589 (N) (allow 590)	**B1**
(b)	correct shape triangle/link given between **R** and the component of the weight down the slope e.g. R = Wcos55	**B1**
	scale given /working shown	**C1**
	resistive force R = 340 (N) (± 20 (N) for scale diagram)	**A2**
(c)	component of W = 589 cos 35	**C1**
	= 482 (N)(allow 480)	**A1**
	(or from triangle and = -P)	
	If the answers are reversed in (b) and (c) then -1	
(d)	resultant force is zero (so no acceleration)/in equilibrium	**B1**
	force up the slope equals force down the slope	**B1**
	(resistive force up slope = component of W down slope scores two)	

2.

(a)(i)	(constant velocity) hence no resultant force/forces add up to zero/ forces balanced / in equilibrium		B1
	friction and (component of) weight down slope = tension (component) up slope		B1
(ii)	**1** 400cos40		C1
	306 (N)	(allow 310)	A1
	2 (400sin40)		
	257 (N)	(allow 260)	A1

(answers to 1 and 2 reversed score 2/3)

3.

(a)	change in velocity / time taken or rate of change of velocity	**B1**
(b)	area under line or using equation $s = (u + v)\,t / 2$ or $s = ut + \frac{1}{2}at^2$	**B1**
	$= (5.6 \times 0.57) / 2$	**B1**
	$= 1.6$ (m)	**A0**
(c)(i)	using $\Delta v = 5.6$ or 5.1 (m s^{-1})	**C1**
	$= [-5.1 - (5.6)] / 20 \times 10^{-3}$	**C1**
	$= (-)535$ (m s^{-2})	**A1**
(c)(ii)	line starts from 0.59 / 0.58 and –5.1	**B1**
	straight line drawn of positive gradient (to the time axis)	**B1**
	same gradient as OA and to the time axis	**B1**

4.

Gradient equated to velocity / speed either explicitly or via a correct example **PLUS ANY FOUR FROM:** **Starts from rest / ends at rest** **Accelerates / velocity increases until a stated time in the range 1 to 2 s or 5 to 6 s.** **Constant speed / maximum speed at 1.5 to 2.5 or 5.5 to 6.5 s.** **Decelerates / velocity decreases from a stated time in the range 2 to 3 s (to 4 s) or from 6 s (to 8 s).** **Zero velocity / speed or at rest at 4.0 s.** **Velocity reverses/ negative after 4.0 s.**	**B1** **B4 MAX 5**	Total of the first five points made by candidate to be marked

Expected Answers	Marks	Additional Guidance
Positive and negative section drawn approximately symmetrical in each half	**M1**	
Correct shape in both sections	**A1**	

5.

speed is distance per unit time / distance / time

velocity is the displacement per unit time / displacement / time

rate of change of velocity / change in velocity per unit time

6.

a)	$t = 2s/(u + v)$ ✓ $t = (v - u)/a$ ✓	2	
b)	$(2s/(u + v) = (v - u)/a)$ to $2as = (uv - u^2 + v^2 - uv)$ ✓ or $2as = v^2 - u^2$	1	not $2as = u^2 - v^2$

7.

(i) 1. speed = distance / time C1
= 22 / 2.4
= 9.2 (9.17) (m s^{-1}) A1

2. velocity = displacement / time C1
= 14 / 2.4
= 5.8 (5.83) (m s^{-1}) A1

(ii) displacement is not equal to the distance B1
displacement is in a straight line
(which is always less than or equal to the total distance) B1

[6]

8.

(a)	force x perpendicular distance from the pivot (to the line of action of the force) (special case –1 if perpendicular is omitted)	M1 A1
(b) (i)	two forces correct two further forces correct	B1 B1
	(Label can be value or name, position judged by eye.)	
(ii)	moments about P / statement of principle of moments	C1
	50g x 0.2 8g x 0.5 + 5g x d (ignore g value used or if g is omitted)	C1 C1
	d = 1.2 hence weight is 0.3 m from B OR: Allow equivalent working backwards	C1 A0
(iii)	move the pivot towards the student or A move the student to the right move the mass to the right increase the value of the mass used increase the weight of the plank increase the strength of the plank increase the length of the plank (any two suitable answers) hence a greater moment can be obtained	M1, M1 A1

9.

(a)(i)	force x perpendicular distance from the force to the pivot / point force x distance from pivot / point scores one	B2
(ii)	(one) force x (perpendicular) distance between the forces	B1
(b)(i)	couple = 25 x 0.3 = 7.5	B1
	unit: Nm (consistent with working)	B1

(ii) Resultant F = 0, Resultant moment ≠ 0, not in equilibrium

10.

(i)	W = 5 x 9.81 = 49 (N) (allow g = 10)	A1
(ii)	arrow acting down (labelled W) drawn approximately halfway from A to B	A1
(iii)	any correct moment	C1
	F x 0.8 = 200 x 0.25 + 49 x 0.4	C1
	F = 87 (N)	A1
(iv)	upward force acts at the hinge	B1
	So F and force at hinge equals weight of table and computer (allow one mark for the upward forces equal the downward forces)	B1

11.

uestion	Expected Answers	Marks
(a)	$T_1 \cos 50 = T_2 \cos 40$ / $T_2 = 392\sin 40$ / $T_1 = 392\sin 50$	
	$T_1^2 + T_2^2 = (392)^2$ / $T_1\sin 50 + T_2\sin 40 = 392$ / correct vector triangle drawn and labelled / scale given for triangle	C2
	$T_1 = 300$ (N)	A1
	$T_2 = 252$ (N)	A1
(b)	Method cannot be used as system not in equilibrium	B1
	Upward forces are greater than the downward forces / resultant force is not zero / cannot form a closed triangle	
	OR	
	Stretching the ropes more gives greater upward force (than downward)	B1
		Total: 6

12.

a		Either: Resolve horizontally F = T cos60 Resolve vertically 2000 = Tsin60 F = 1155 (N) Or correct triangle of forces correct trig. statement F = 1155 (N) Or correct scale diagram Scale given F = 1155 (N)	 C1 C1 A1 C1 C1 A1 C1 C1 A1	 Allow answers to two SF (1200) Basic shape (30, 60 and 90), two forces labelled (arrows not needed). Give this mark even if 30 and 60 reversed (this gives 3464N and would gain 1 mark max). Allow answers to two SF (1200) As above (for triangle of forces) Allow answers from 1100 to 1200 (N)
b	i	Larger mass, weight, larger angle or larger force F Greater kinetic energy (change) or greater momentum (change) Allow greater speed <u>at the point of collision</u> /where <u>it hits the wall</u> or mass is larger hence force <u>on wall</u> is larger using F=ma	 B1 B1	Not longer cable Do not accept It hits with greater force on its own Do not accept travels with greater acceleration hence force is larger using F=ma.

13.

(i) 12 N down requires 12 N up / net force vertically is zero / 6 sin30 **C1**
(3 N from right string hence) 9 N needed from T **A1**

(ii) T cos30 = 9 / T = 12 cos30 / or correct triangle of forces used (possible ECF) **C1**

T = 10.4 (N) (Allow ± 0.2N if scale diagram used) **A1**

14.

$S = ut+0.5at^2$

$S = 0.5at^2 = 0.5 \times 9.81 \times 1.75^2$

$S = 15m$

15.

(a)	$s = {}^1/_2 \times 9.8 \times (4.0)^2$ ✓ m = 78.4 m ✓e	2
(b)	resistive/frictional forces✓ so acceleration less / time too long ✓ (or takes time for sound to come back✓, so time actually less ✓ than 4s)	2

16.

$s = {}^1/_2at^2$ $t^2 = (2 \times 0.15)/9.8$ ✓✓ t = 0.18 s ✓ g = 10 gives 0.17 s	3	using t = 0.2 s to find s = 0.196 m ✓✓ then explained ✓

17.

(a)	$t^2 = (2 \times 6.80) / 9.81$ ✓$_m$ t = **1.18** (s) ✓$_e$ **3 s.f. only**	2	using g = 10 accept t = 1.17 here
(b)	v = 6.80 / Δt ✓$_m$ correct evaluation ✓$_e$ (m s^{-1}) (Δt = 1.20 – (a)) *(note: erroneous answer for (a) can gain ✓✓ in (b))*	2	poss. evaluations: 340 (using t = 1.18) 309 (using t = 1.178) 227 (using t = 1.17) 200 (using t = 1.166)

18.

(a)	$114 \times 10^3 / (60 \times 60)$ = **31.7** ✓$_e$ (**calculator** value)	1	
(b)	a = 31.7 /8.4 ✓$_m$ = **3.8** (m s^{-2}) ✓$_e$	2	

19.

(a) (i)	(**using** $s = {}^1/_2at^2$) 3.0 = ½ x 9.8x t^2 ✓$_m$ t = 0.78 ✓$_e$ (s)	2	✓$_e$ calculator value ora
(ii)	(using v = u + at) = 9.8 x 0.78 ✓$_m$ = 7.6 ✓$_e$ (m s^{-1}) (v^2 = 2 x 9.8 x 3.0 gives v = 7.7 m s^{-1})	2	9.8x0.8 = 7.8 (m s^{-1})

20.

(i)	W vertically down at G	**B1**
	Force at O vertical	**B1**
(ii)	$V \times 0.9 \times \cos 60 = W \times 0.35 \times \cos 60$	**B1**
	$V = (25 \times 0.35) / 0.9$	**B1**
	$= 9.7(22)$ (N)	**A0**
(iii)	total force is zero stated or implied / 25 – 9.7	**C1**
	force at hinge = 15.3 (N)	**A1**
	(or may take moments about G or V)	